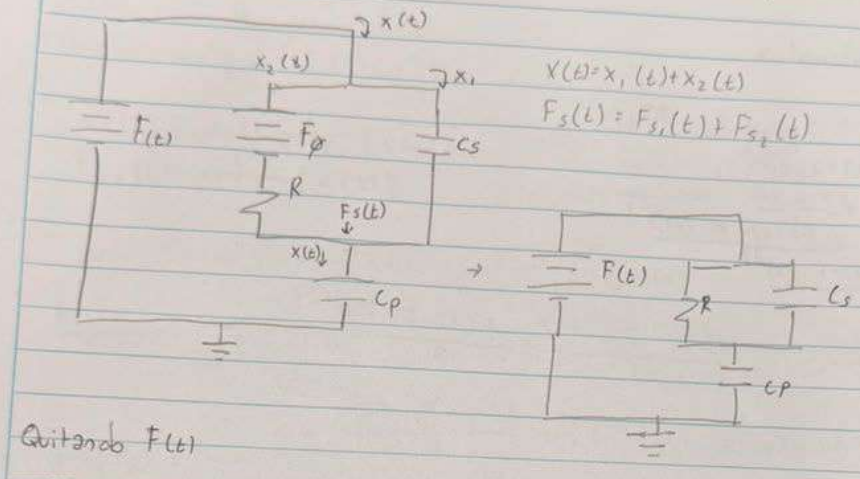
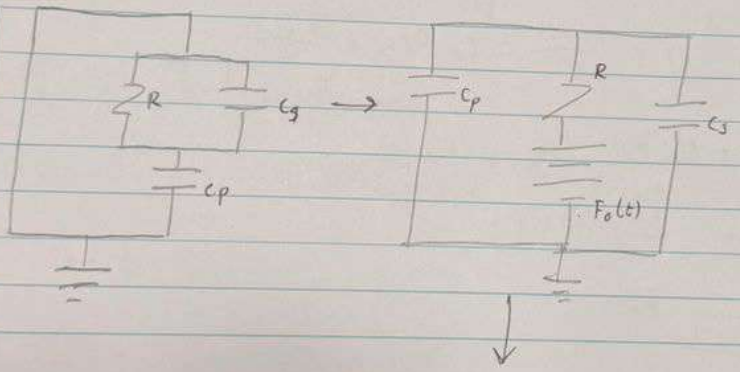
Practica 3: Modelado y control del sistema musculoesqueletico

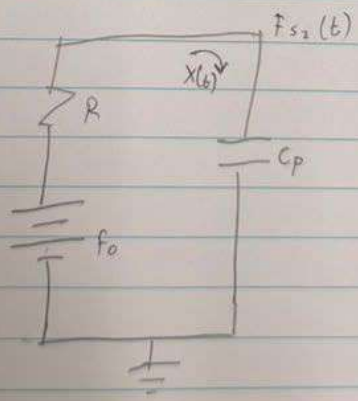


$X(t) = X_1(t) + X_2(t)$

$F_s(t) = F_{s_1}(t) + F_{s_2}(t)$

Quitando F(t)





Analisis matematico

Circuito 1: Apagando $F_o(t)$

$X(t) = X_1(t) + X_2(t)$

$x(t) = C_p \frac{dF_{s_1}(t)}{dt}$

$X_1(t) = C_s \frac{d[F(t) - F_{s_1}(t)]}{dt}$

$X_2(t) = \frac{F(t) - F_{s_1}(t)}{R}$

$$C_p \frac{dF_{s_1}(t)}{dt} = C_s \frac{d[F(t) - F_{s_1}(t)]}{dt} + \frac{F(t) - F_{s_1}(t)}{R}$$

$$C_p s F_{s_1}(s) = C_s [F_{s_1}(s) - F_{s_1}(s)] + \frac{F(s) - F_{s_1}(s)}{R}$$

$$C_p s F_{s_1}(s) + C_s F_{s_1}(s) + \frac{F_{s_1}(s)}{R} = C_s F(s) + \frac{F(s)}{R}$$

$$(C_p s + C_s + \frac{1}{R}) F_{s_1}(s) = C_s + \frac{1}{R} F(s)$$

$$\frac{R(C_p + C_s)s + 1}{R} F_{s_1}(s) = \frac{C_s R_s + 1}{R} F(s)$$

$$F_{s_1}(s) = \frac{\frac{C_s R + 1}{R}}{\frac{R(C_p + C_s)s + 1}{R}} F(s)$$

$$F_{s_1}(s) = \frac{C_s R_s + 1}{R(C_p + C_s)}$$

Circuito 2. Apagando $F(t)$

Recordando que $F_a(t) = \alpha F(t)$

$$-\alpha F(t) = R_x(t) + \frac{1}{C_p + C_s}\int x(t)\,dt \quad ; \quad -\alpha F(s) = R_x(s) + \frac{x(s)}{(C_p+C_s)s}$$

$$F_{s_2}(t) = \frac{1}{C_p + C_s}\int x(t)\,dt \quad ; \quad F_{s_2}(s) = \frac{x(s)}{(C_p+C_s)s}$$

$$F(s) = -\frac{1}{\alpha}\left(R + \frac{1}{(C_p+C_s)s}\right)x(s) = -\frac{1}{\alpha}\,\frac{R(C_p+C_s)s+1}{(C_p+C_s)s}\,x(s)$$

$$\frac{F_{s_2}(s)}{F(t)} = -\frac{1}{\alpha}\,\frac{\frac{x(s)}{(C_p+C_s)s}}{\frac{R(C_p+C_s)s+1}{(C_p+C_s)s}}\,x(s) = -\frac{\alpha}{R(C_p+C_s)s+1}$$

$$F_{s_2}(s) = -\frac{(a\,F(s))}{R(C_p+C_s)s+1}$$

Funcion de transferencia

$$F_s(s) = F_{s_1} + F_{s_2}(s) = \frac{C_s R_s + 1}{R(C_p+C_s)s+1}\,F(s) - \frac{a\,F(s)}{R(C_p+C_s)s+1}$$

$$F_s(s) = \frac{C_s R_s + 1 - a}{R(C_p+C_s)s+1}\,F(s)$$

$$\frac{F_s(s)}{F(s)} = \frac{C_s R_s + 1 - a}{R(C_p+C_s)s+1}$$

Error en estado estacionario

$$e(s) = \lim_{s \to 0} s \; F(s)\left(1 - \frac{F_s(s)}{F(s)}\right) = \lim_{s \to 0} s \frac{1}{s}\left(1 - \frac{RC_s s + 1 - \alpha}{(RC_p + RC_s)s + 1}\right) = \alpha$$

Estabilidad en lazo abierto

$$(RC_p + RC_s)s + 1 = 0$$

$$\lambda = \frac{1}{RC_p + RC_s}$$

El sistema es estable.